



English, Economics, Politics, and Philosophy

Unit 2 Topics in Business

Lesson 2.5: The Game of the Market

Pertinent Topics:

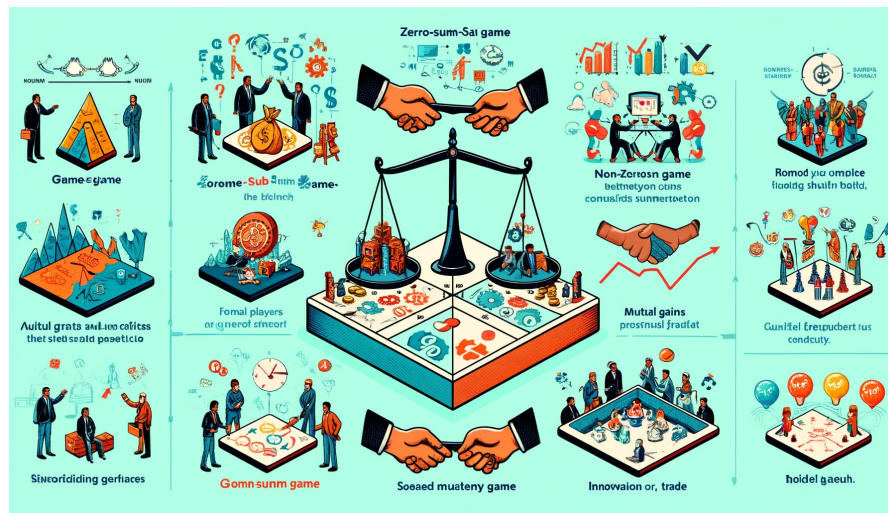
- Evolutionary theory for Economics
Evolve
- Analogy from game theory

Intro & Recap:

In previous tutorials, we learned about different theories for rationality in the context of game theory. We also learned about expected value theory according to Causal Decision Theory and Evidential Decision Theory. Indeed, the insights from game theory are not only applicable to our decision-making process, they have extended a large body of literature in philosophy as well.

MINDS ON

Thus far the game theory models with which we engage are called the two player's **non-zero-sum** (i.e., possible for mutual benefits) game. A game is said to be zero-sum when one party's gain is exactly balanced by another party's loss. Think about our economy as a societal game, is it zero-sum (that is, is it merely a winner and loser game)? Also, do you think n (finitely many) player game is possible?



Aa: Yes I think it is non-zero-sum (winner loser game). Because when you buy something both you and the seller gain happiness.
 Ab: I think there is limitation for the numbers player in a game

price: cost + profit + tax (with tax price gone up)
law of demand tells us that the quantity decreases
overall profit decreases/ overall utility decreases

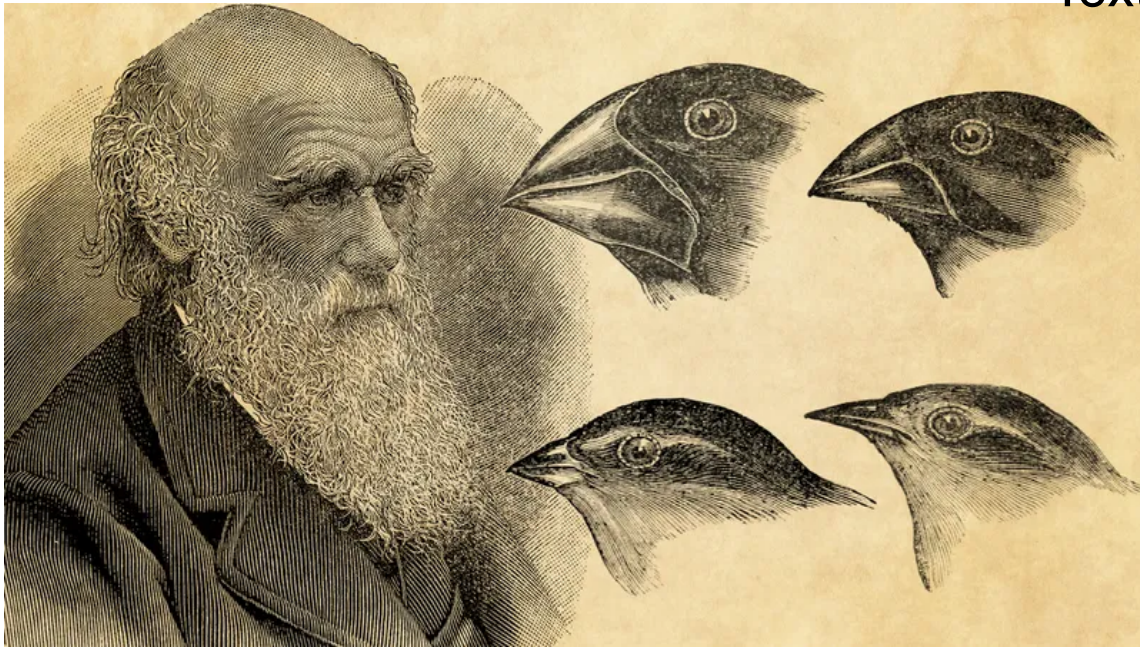
I. Evolutionary theory for economics
there is an excess burden

Charles Darwin — The Father of Evolutionary Biology

agnosticism

agnostic (confused)

Text



Charles Darwin (1809-1882) was an English **naturalist** whose scientific theory of **evolution by natural selection** became the foundation of modern evolutionary studies. He developed his theory of evolution through careful observation and study.

C.f. **Invisible Hand**
During his voyage on the HMS Beagle, he collected many specimens and noticed **variations among species in different environments**. His reading of **Thomas Malthus's work on population growth and competition** inspired him to think **about natural selection** (because of Malthus emphasis on the incompatibility between the growth rate of population and finite resources). This idea, where the fittest survive and

Survival the fittest

reproduce, became the cornerstone of his theory. Darwin was also influenced by Adam Smith's economic ideas, especially the concept of the invisible hand, which he saw reflected in nature's way of regulating itself.

Darwin's ideas about evolution influenced economists in the Austrian School, like Joseph Schumpeter and Friedrich Hayek. Schumpeter's idea of "creative destruction," where new innovations (what Schumpeter called industrial mutation) replace old industries, mirrors natural selection. Hayek saw the economy as an organic system that evolves over time, rather than a machine that can be controlled. Both economists believed that the economy works best when it can adapt and change freely, similar to how species evolve in nature. This perspective highlights the impact of Darwin's theory on their economic thinking.

survival the fittest (adaptive adaptability)
We can contrast the works of Austrian School with Keynesian School. For the followers of Keynes, the economy can and should be engineered by the government. This is because even if the planning goes wrong, the emotions of the population have to be placed as the priority (politically). And so his subsequent development of income expenditure model.

thus far, we have the analogy from biology to economy

II. Analogy from game theory



Well, what can we say about these economists' theories using insights from game theory? On individual level, decision making process seems to be one player game in causal decision's sense. That is, without prior knowledge, we plan individually according the predicted outcomes caused by our actions: this is by the way call Praxeology popularized by Ludwig von Mises (whom we will learn later). On societal level it is a n-player game where there are finitely many participants (us). Still, we can attempt to 'optimize' the outcome through perhaps Pareto Optimum, but is it always desirable for individuals? Are we (government) justified in planning for us?

Pareto Optimum may not be the best individually

Let's group Hayek, Schumpeter, and Smith together. For them, **or plan the society** is it possible to 'calculate' n-player game?

NO they don't think it that way. it is rather impossible because (1 Hayek) individual preferences are not the collective preferences, (2 Schumpeter) industrial mutation is not predictable, (3 Smith) invisible hand is invisible

What about Keynes and Marx?

Yes. Keynes economy can be engineered government spending will stimulate economics activity and government investment is a part of production; Marx thinks we can organize labours into "from each according to each ability, to each ability to each needs"

Assignment: after the lesson, have your thoughts on the possibility of n-player game change especially when it is analogously used for **social planning**? If so, explain. If not, defend your stance. **from the perspective of the government you plan the economy and people's affairs**

